

Solution Requirements

Date	30 June 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (credit officers, analysts) securely log in using unique credentials; optional two-factor authentication for added security.	High
2	Authorization levels	Role-based access: Applicants can submit data; Credit Officers can view predictions; Admins can manage models and user accounts.	High

3	External interfaces	Integration with IBM Cloud for deployment; optional APIs for connecting to bank databases or credit history services.	Medium
4	Transactions processing	Applicant data is collected, validated, and processed through ML models; predictions are stored and displayed in real time.	High
5	Reporting	System generates dashboards showing approval rates, risk scores, and model performance metrics; exportable as CSV or PDF.	Medium
6	Business rules, etc.	Loan approval logic based on model output (e.g., approve if probability of repayment > threshold); flag high-risk applicants for manual review.	High
7	Compliance to laws or regulations	Adheres to data privacy standards (GDPR-equivalent), ensures secure handling of applicant financial data.	High
8	<i>Other</i>	Includes audit logging for all predictions and user actions to ensure transparency and traceability.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process and return predictions quickly for real-time decision-making.	<i>Page loads and predictions in < 2 seconds)</i>
2	Scalability	Should handle increasing numbers of applicants and concurrent users without performance degradation	Supports up to 10,000 concurrent users
3	Security & Data Privacy	All applicant data encrypted in transit and at rest; secure authentication and access control.	End-to-end AES-256 encryption
4	Reliability & Availability	System should remain operational with minimal downtime; fault-tolerant architecture.	99.9% uptime per month
5	Usability & Accessibility	Intuitive UI for applicants and officers; accessible across devices and compliant with accessibility standards.	WCAG 2.1 AA compliance
6	<i>Other</i>	Codebase modular and well-documented; easily deployable on other cloud platforms if needed.	Deployment-ready on IBM Cloud and portable to AWS/Azure